B.Tech. V Semester (Computer Engg.) Examinations, JMI

Session 2021-22

Time: 1 hrs. **Computer Networks (CEN - 505)** **Sessional-II** Max. Marks: 15

Note: Attempt All Questions

1 CO3	In a TDM medium access control bus LAN, each station is assigned one time slot per cycle for transmission. Assume that the length of each time slot is the time to transmit 100 bits plus the end to end propagation delay. Assume a propagation speed of $2 * 10^8$m/sec. The length of the LAN is 10 km with a bandwidth of 1 Mbps. Find the maximum number of stations that can be allowed in the LAN so that the throughput of each station can be 1/3 Mbps.	3
2 CO3	Station A needs to send a message of 12 packets to station B using a sliding window (window size 4) and Go-back-N error control strategy. All packets are ready and immediately available for transmission. If every 4th packet that A transmits get lost (but no acknowledgement form B ever get lost), then what is the number of packets that A will transmit for sending the message to B?	3
3 CO4	What is the use of virtual channel sensing in CSMA/CA? In wireless LAN, the point coordination function and distributed coordination function can coexist within one cell. Explain how it is possible to have central control and distributed control operating at the same time.	4
4 CO5	Discuss the problem of count to infinity associated with distance vector routing technique. For the following figure, the network uses a Distance Vector Routing protocol. The following vectors have just come into router C: from B: (5,0,8,15,6,2); from D: (18,12,6,0,9,12); and from E: (8,7,3,9,0,4). The measured delays to B, D and E are 8, 4 and 6, respectively. What is C's new routing table? Give both the outgoing line to use and the expected delay.	4

